

Article

Arbuscular Mycorrhizal Fungi Inoculation and Water Regime Effects on Seedling P Uptake by Rice and Pearl Millet

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Abstract: Mycorrhizal-mediated seedling establishment may reduce dependency on chemical fertilizers, but the effectiveness of infection for growth may differ depending on species with different eco-physiological adaptations. The infection of arbuscular mycorrhizal fungi (AMF) and P uptake were compared between rice (*Oryza sativa* L.) (Koshihikari (rice_k), Togo4 (rice_t)), and pearl millet (*Pennisetum glaucum* (L.) R. Br.) (ICMB89111 (millet₈₉₁), ICMB95444 (millet₉₅₄)) seedlings (i) in response to three different commercial AMF inoculants of *Rhizoglyphus irregularis* (popular inoculant Dr. Kinkon (I₁); two new inoculants Rootella P (I₂) and Rootella F (I₃)) in comparison with indigenous AMF from Andosol upland and paddy topsoils (Exp. 1–2 as the inoculant experiments) and (ii) across different water regimes from upland to flooded lowland conditions for I₁ inoculant (Exp. 3–4 as the water regime experiments). The new inoculants I₂ and I₃ with higher propagule numbers showed a higher infection rate than the control seedlings in both rice and pearl millet, with a tendency for slower leaf development and no seedling growth enhancement. I₁ inoculant had more significant positive effects on the root transversal area and shoot growth parameters than the control. The infection rates of all three inoculants were lower than the indigenous AMF from upland Andosol in rice and pearl millet, in which a higher infection rate led to higher P uptake found in millet₉₅₄. I₁ inoculant increased the infection rate in pearl millet and rice but had no clear indication of interaction with water regimes. A higher infection rate led to higher P uptake and shoot dry weight in pearl millet but not in rice with higher root length density. This study provided the significance of inoculants for seedling establishment and highlighted more mycorrhizal-mediated P uptake in pearl millet than in rice.

Keywords: arbuscular mycorrhizal fungi; inoculation; pearl millet (*Pennisetum glaucum* (L.) R. Br.); rice (*Oryza sativa* L.); water regime



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1. Introduction

The mycorrhizal-mediated seedling establishment may reduce dependency on chemical fertilizers, but the effectiveness of inoculants and their interaction with water regimes are poorly understood. Arbuscular mycorrhizal fungi (AMF) may help plant establishment by providing nutrients and water supply to the host crops [1]. Infection rates may differ between commercial inoculants and soil-indigenous AMF, and there may be an optimal level of water regimes for effective inoculation.

AMF colonization is known to start from the first week after inoculation; germinated spores could develop external hyphae in 1 day, internal hyphae after the entry point in 2 days, and then arbuscules and vesicles in ~4 days [2]. Inoculation with AMF in a seedling

bed improved plant growth in the seedling stage (3–4 weeks old as in rice and pearl millet seedlings (knowledgebank.irri.org, accessed on 19 March 2025; [3])), which led to superior growth after transplanting in the fields, higher nutrient uptake, and higher yield [4]. The infection rate of 4-week-old rice seedlings reached 40% at maximum [4], which is usually much lower and varies depending on the various factors (e.g., water regimes, types of inoculants, the number of inoculant spores or propagules, and P availability). For example, an upland nursery (a 60% water holding capacity) had a three times higher infection rate than a flooded nursery (a 3 cm water depth) [4]. Auge reviewed inoculation effects under drought in many plants, showing positive responses such as growth, plant water status, and stomata conductance in some cases [5]. However, neither rice nor pearl millet was covered.

AMF commonly resides in the soil as indigenous. Commercial AMF inoculants with some AMF species are available. Many commercial inoculants have been used and reported internationally; for example, Salomon et al. [6] reported nearly 30 inoculants. *Rhizoglossum irregulare*, formerly known as *Rhizophagus irregularis* or *Glomus intraradices* [7], is the model AMF due to its ability to produce high P uptake [8] and propagation in vitro. In Japan, only a few products have been used, among which Dr. Kinkon (*Glomus* sp. strain R-10) (Idemitsu Kosan Co., Ltd., Tokyo, Japan) (<https://www.idemitsu.com/jp/content/100038434.pdf>, accessed on 19 March 2025) has been the most popular product in Japan for over 20 years; it was tested on wild legume [9], onion [10], and soybean [11]. Recently, sets of new products from Israel were introduced in Japan as Rootella with *R. irregularis* (Groundwork BioAg Co., Ltd., Moshav Mazor, Israel, accessed on 27 March 2025), which claimed to have positive effects on growth, yield, nutrient uptake, and response to stresses to many crop species. Several types of Rootella products are available (e.g., Rootella P (strain P-type), Rootella F (strain F type)), whose propagule numbers are much higher than Dr. Kinkon and may increase infection further. It should not be neglected that natural soils in fields could have large numbers of indigenous AMF that would cause positive plant growth, as reported in Andosol's case by Solaiman and Hirata [12]. It should also be kept in mind that inoculants do not always have positive effects on crop growth [13,14].

Here, this study aimed to make a preliminary investigation of (1) the effects of three different commercial AMF inoculants and indigenous AMF from Andosol upland and paddy soils and (2) the effects of AMF inoculation under different water regimes on the infection and growth of rice and pearl millet at seedling stages. The new inoculant with higher propagule numbers may significantly affect crops and there may be an interaction with water regimes. In addition, rice and pearl millet with different root systems may respond differently to enhanced infection rates.

2. Materials and Methods

Three experiments (Exp. 1–3) were simultaneously conducted from June to July 2020, and another experiment (Exp. 4) was conducted from May to June 2021 in the greenhouse at the Institute for Sustainable Agro-ecosystem Services (ISAS), The University of Tokyo, Nishitokyo, Japan (35°43' N, 139°32' E). All the experiments were conducted during the seedling stage in cell trays, which were rotated weekly: Exp. 1 with three inoculant types (2020), Exp. 2 with indigenous AMF from two soil types (2020), Exp. 3 with four water regimes with inoculant (2020), and Exp. 4 with five water regimes with inoculant and control (2021).

2.1. Plant Materials

Two genotypes (hybrid 'Togo4', 'Koshihikari') of rice (*Oryza sativa* L.) and two (ICMB89111, ICMB95444) of pearl millet (*Pennisetum glaucum* (L.) R. Br.) were used. Koshihikari (rice_k) is an improved lowland genotype of *O. sativa* ssp. *japonica* and Togo4 (rice_t)

is a high-yielding, good-eating quality hybrid rice with a rice_k background, developed by the Research Institute of Rice Production & Technology Co., Ltd., Toyoake, Japan. ICMB89111 (millet₈₉₁) is a high-tillering inbred genotype, and ICMB95444 (millet₉₅₄) is a hybrid genotype, both of which were developed by the International Crop Research Institute for Semi-Arid Tropics (ICRISAT), Hyderabad, India.

2.2. Experimental Design

2.2.1. Three Inoculant Types (Exp. 1)

Before sowing, nursery soil containing 29.8% of Akadama (Sharaka Co., Kanuma, Tochigi, Japan), 44.7% of Kanuma (Tomiya Engei, Tenri, Nara, Japan), 17.9% of Ezo Sand (Plantation Iwamoto Co., Hokota, Ibaraki, Japan), and 7.4% of Baido soil (Akagiengai Co., Isesaki, Gunma, Japan) was autoclaved at 121 °C for 30 min to eliminate all the microorganisms, including AMF, which might exist in the nursery soils. Then, the soil was subsampled and oven-dried for three days at 100 °C to determine its gravimetric water content (GWC, %). With no fertilization added, the autoclaved soil was mixed with one of the three commercial inoculants, Dr. Kinkon (R10; I₁), Rootella P (strain P-type; I₂), or Rootella F (strain F-type; I₃), at 10 g/kg of nursery soil based on the products' recommendation. Dr. Kinkon (I₁) contained R10 spores, external hyphae, and root fragments with the crystalline-silica carrier and the most probable number of 14 propagules/g with 21 OTUs found in *R. irregularis* [15]. According to the manufacturer, Rootella P (I₂) has at least 2500 viable propagules/g with 88% of clay and 12% of the active ingredient, and Rootella F (I₃) has at least 20,800 viable propagules/g with 92% of vermiculite and 8% of the soil amendment ingredient.

Seeds were soaked in cups on 29 June and on 2 July, sowed one per cell (~2 × 2 × 2 cm) in a 100-cell nursery tray (per inoculant treatment, a total of 20 plants for each genotype with 20 additional rice_k as border plants). Each nursery tray with 1.4 kg of soil was placed inside a flat tray, about 1 cm taller than the nursery tray. With an electric scale, the whole trays were weighted and watered daily to maintain a GWC of around 50% based on daily water loss until harvesting on 23 July. Daily minimum and maximum temperatures inside the greenhouse during treatment were 21 and 33 °C.

2.2.2. Indigenous AMF in Two Soil Types (Exp. 2)

Topsoil (0–10 cm) from the paddy lowland (PD) and upland (UP) fields of ISAS in the spring of 2020 was collected, sieved by 0.5 cm mesh to remove debris, and used without autoclaving and adding any fertilizer. The soil chemical properties were as below: the pH (H₂O) was 6.6 and 6.6, the total C was 10.6 and 10.4%, the inorganic N was 3.46 and 0.53 mg/100 g, the P was 1.30 and 2.99 mg/100 g, and the K was 21.1 and 49.3 mg/100 g, in PD and UP, respectively. The soils were subsampled and oven-dried for three days at 100 °C to determine their GWC. Growth conditions and dates in Exp. 2 were the same as in Exp. 1.

2.2.3. Water Regimes (Exp. 3 and Exp. 4)

Like Exp. 1, nursery soil in Exp. 3 (29.8% of Akadama, 44.7% of Kanuma, 17.9% of Ezo Sand, and 7.4% of Baido soil) was autoclaved at 121 °C for 30 min and later oven-dried for three days at 100 °C to determine the GWC (%). The autoclaved soil was mixed with I₁ for 10 g/kg of nursery soil based on the products' recommendation without adding any fertilization. Until 12 July, growth conditions and dates in Exp. 3 were the same as in Exp. 1. From 13 to 23 July, four water regimes were imposed and monitored daily: water irrigated and maintained at 1 cm above the soil surface (flooded, FL), water held at a similar amount as before treatment (well irrigated, W100), water reduced by 50% of water amount added in W100 (50% well irrigated, W50), and water reduced by 25% of water amount added in

W100 (25% well irrigated, W25). During treatment, whole trays were weighed, and the water amount was added based on the water loss each day. Soil GWC from 13 to 23 July was highest in W100 (53%), W50 (47%), and W25 (41%), respectively. In FL, 30–40% of pearl millet remained for harvest.

In Exp. 4, Kubota nursery soil with N-P-K 0.23–0.32–0.23 g/kg of soil (KN-1U, Kubota Co., Osaka, Japan) was autoclaved at 121 °C for 30 min and oven-dried for three days at 100 °C to determine the GWC (%). The autoclaved soil was added to I₁ at 7 g/kg in the inoculation treatment but not in the control treatment. Seeds were soaked on 14 May for pearl millet (again on 16 May due to its poor germination) and 17 May for rice and sowed at one seed per cell ($\sim 5 \times 5 \times 5$ cm) in the 50-cell nursery tray (per treatment, a total of 10 plants in each genotype with an additional 5 for millet₈₉₁ and rice₁) on 18 May. Each nursery tray was placed inside a flat tray, which was taller by about 1 cm than the nursery tray as in Exp. 3. The whole trays were weighted bidaily with an electric scale, and water amounts were added based on the water loss each day to maintain a GWC of around 30%. To replicate field conditions, from 2 to 16 June, five water regimes were imposed and monitored daily: water irrigated and held at 4 cm above soil surface by placing the tray into a 60 L tank (flooded, FL), water added to saturated (soil saturated, SS), water maintained at the similar amount to that of before treatment (well irrigated, W100), water reduced by 50% of the water amount added in W100 (50% well irrigated, W50), and water reduced by 25% of the water amount added in W100 but bidaily irrigated (25% well irrigated, W25). During the treatment, whole trays were weighed, and the water amount was added based on the water loss each day. Soil GWC from 2 to 16 June was highest in SS (47%), followed by W100 (31%), W50 (28%), and W25 (27%), respectively. Before treatment, a few plants in each genotype were collected for the infection rate, and after treatment, the survival rates of millet₈₉₁ and millet₉₅₄ in FL and SS were 50–60% in the control and 40–50% in the inoculated plants. Daily minimum and maximum temperatures inside the greenhouse during treatment were 17 and 44 °C.

2.3. Measurements

2.3.1. Shoot Growth

Shoot growth parameters such as tiller number (TN), plant height (PH; cm), leaf age (LA), and shoot dry weight (SDW; g/plant) with 4–5 plants per treatment were collected on 23 July 2020 ($n = 4$ –5) and 16 June 2021 ($n = 4$). LA was recorded from the first leaf. For SDW, plants were collected and oven-dried for three days at 80 °C.

2.3.2. Shoot Phosphorus Concentration and Uptake

The dry shoots of one or two plants were mixed and ground into powder using a fine mill (Heiko sample mill, TI 300, Fujiwara Seisakusho, Tokyo, Japan) and digested by the MARS 6 microwave (CEM, Matthews, NC, USA) following the modified method note on plant tissue digestion (<https://cem.com/ja/digestion-of-plant-tissue-mars-6>, accessed on 19 March 2025). The digestion was started by weighting the sample ~ 0.01 –0.1 g by a four-decimal scale before placing it into a digestion vessel, adding 2.5 mL of 68% concentrated nitric acid (HNO₃) and 0.5 mL of 30% hydrogen peroxide (H₂O₂) in the draft and tightening the lid of the digestion vessels. The microwave was set as follows: Stage: 1; Temp (°C): 220; Ramp (mm: ss): 30:00; Hold (mm: ss): 30:00; Pressure (psi): 800; Power (W): 1200; Stirring: off. After cooling for 20–30 min, the vessel's solution was transferred to a 10 mL tube by rinsing with 2 mL of distilled water twice before filling it up to 5 mL. The solution was mixed and filtrated by a minisart syringe filter (pore size, 0.22 μ m; Sartorius Co., Goettingen, Germany) up to 4 mL. Standard solutions were prepared as 0, 10, and 50 ppm for phosphorus (P) in three separate 50 mL tubes. The concentration of each standard

solution and sample was calibrated and measured by inductively coupled plasma optical emission spectrophotometers (ICP-OES, SPS3000, Hitachi High-Technologies, Tokyo, Japan) following its operation manual. Finally, the concentration (ppm) with dry weight was calculated for nutrient concentration (%) and uptake (mg/plant) ($n = 3$).

2.3.3. Root Growth

Root length density (RLD; cm/cm^3) in 2020 ($n = 5$) and 2021 ($n = 4$) was measured from all roots, including the tray bottom. After washing with tap water, roots were immersed in 50% ethanol until measurement. All roots were arranged on a transparent plate and scanned with an Epson Expression 11000XL scanner in professional mode, with positive film, and an 8-bit grayscale at 600 dpi [16]. The images were then analyzed by WinRhizo Pro (Regent Instruments, Quebec, QC, Canada). After scanning, roots at 2–3 cm from the base (one per replication) were cross-sectioned and examined under a phase contrast microscope (BX51, Olympus, Tokyo, Japan). CellSens standard software ver 4.2 (Olympus) was used to capture the microscopic images. Root transversal area (RTA; μm^2) was measured by the polygon tool in ImageJ 1.51t (NIH, Bethesda, MD, USA) [17].

2.3.4. Root AMF Infection

A representative sample of roots (0.3–0.8 mm in diameter) in 50% ethanol was stained with the trypan blue method [18]. First, 5–10 pieces of ~2 cm roots were cleared with 10% (w/v) KOH by boiling at 110 °C for 15 min, rinsed once with water, soaked in 2% (w/v) hydrochloric acid (HCl) at room temperature for ~5 min and dyed with trypan blue solution (0.05% in lactic acid) by boiling at 90 °C for ~5 min. Then, after rinsing thrice with water, roots were stored in lactoglycerol with a ratio of lactic acid/glycerol/water, 8:1:1 ($v/v/v$). Finally, AMF-infected roots were quantified with a modified gridline intersection method [19]. Three roots per replication were placed parallel in a slide glass with a gridline of 1 mm $\times$ 1 mm with cover glass and observed at 100 $\times$ or 200 $\times$ magnification under a phase-contrast microscope (BX51, Olympus, Hicksville, NY, USA). In total, nine segments of 2 cm roots per 4–5 replications, or 240–300 intersections per treatment, were observed.

At each intersection, the presence or absence of AMF structures was scored for arbuscules, vesicles, and intraradical hyphae. Arbuscules are plant-like structures serving as nutrient exchange sites. Vesicles are oval or circular structures acting as lipid storage rooms, and intraradical hyphae are long, thin, fungal filaments. Arbuscules and vesicles were counted separately, while intraradical hyphae were counted if other mycorrhizal structures were present due to either arbuscules or vesicles implying the presence of hyphae (Figure S1). The mycorrhizal infection rate (or intraradical hyphal infection rate; $M\%$) was calculated by dividing the total number of infected root intersections by the total number of observed root intersections ($n \approx 60$). The arbuscular ($A\%$) and vesicular ($V\%$) infection rates were determined by dividing the number of arbuscule and vesicle intersections by the total number of observed root intersections.

$$M\% = \frac{\text{Total number of infected root intersections}}{\text{Total number of observed root intersections}} \times 100$$

$$A\% = \frac{\text{Total number of intersections with arbuscules}}{\text{Total number of observed root intersections}} \times 100$$

$$V\% = \frac{\text{Total number of intersections with vesicles}}{\text{Total number of observed root intersections}} \times 100$$

2.4. Statistical Analysis

Data were analyzed using GenStat 21.1 (VSNi, Hemel, Hempstead, UK). General ANOVA and multiple comparisons (Tukey's test) (significance set at $p < 0.05$) were used to assess the effects of water (4–5 water regimes), inoculation (C, I), genotype (millet₈₉₁, millet₉₅₄, rice_t, and rice_k), inoculants (I₁, I₂, and I₃), and soils (UP and PD) on infection rates (M%, A%, and V%), P concentration (P%), and uptake and plant growth parameters (SDW, PH, TN, RLD, and RTA). Multiple comparisons for the interactive effects were assessed by the least significant difference (significance set at $p < 0.05$).

3. Results

3.1. Exp. 1 Inoculation Types

Inoculation affected the infection rate (M%), root growth (RLD and RTA), and shoot growth (SDW, PH, and LA) (Table 1). The M% in I₂ was the highest, followed by I₃. I₂ had the lowest LA and the least SDW. The M% was not significantly different between I₁ and control, while I₁ had the highest RLD, RTA, SDW, and PH. Rice_k had a higher M% than millet₈₉₁ and rice_t, and root and shoot growth parameters were generally larger in rice than pearl millet. The inoculant type interacted with genotypes as I₁ enhanced the RLD and SDW for rice_k, and I₂ decreased the RLD and SDW for rice_k (Table 2).

Table 1. Effects of inoculation and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rates in (%), root length density (RLD, cm/cm³), root transversal area (RTA, ×10⁴ μm²), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), plant height (PH, cm), and leaf age (LA) with control (C) and inoculants Dr. Kinkon (I₁) and Rootella P (I₂) and F (I₃) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 1).

Parameters	M%	A%	RLD	RTA	P%	P Uptake	SDW	PH	LA
Inoculation (I)	***	NS	***	*	NS	NS	***	NS	***
C	0.0 ± 0.0 a	0.0 ± 0.0	27 ± 3 ab	9.6 ± 0.9 ab	0.146 ± 0.018	0.017 ± 0.003	0.011 ± 0.000 ab	11.4 ± 0.2	3.8 ± 0.0 b
I ₁	2.4 ± 1.1 a	0.0 ± 0.0	31 ± 5 b	12.5 ± 1.4 b	0.173 ± 0.027	0.023 ± 0.005	0.013 ± 0.000 b	12.9 ± 0.3	3.6 ± 0.1 ab
I ₂	13.1 ± 3.7 b	0.0 ± 0.1	19 ± 3 a	10.5 ± 1.3 ab	0.149 ± 0.009	0.014 ± 0.004	0.008 ± 0.001 a	12.2 ± 0.7	3.4 ± 0.1 a
I ₃	7.5 ± 2.6 ab	0.1 ± 0.0	21 ± 2 a	8.4 ± 0.8 a	0.126 ± 0.008	0.016 ± 0.003	0.012 ± 0.002 b	12.2 ± 0.8	3.5 ± 0.1 a
Genotype (G)	*	NS	**	**	NS	***	**	***	***
millet ₈₉₁	2.2 ± 0.9 a	0.0 ± 0.0	17 ± 2 a	9.1 ± 1.3 a	0.135 ± 0.018	0.010 ± 0.001 a	0.006 ± 0.002 a	6.2 ± 1.3 a	3.3 ± 0.1 a
millet ₉₅₄	5.8 ± 1.9 ab	0.1 ± 0.1	14 ± 3 a	7.9 ± 0.8 a	0.140 ± 0.012	0.007 ± 0.001 a	0.004 ± 0.002 a	5.2 ± 1.7 a	3.3 ± 0.1 a
rice _k	11.4 ± 4.1 b	0.0 ± 0.0	28 ± 2 b	10.5 ± 0.9 ab	0.173 ± 0.027	0.024 ± 0.004 b	0.013 ± 0.002 b	17.2 ± 1.6 b	3.9 ± 0.1 b
rice _t	3.6 ± 1.8 a	0.0 ± 0.0	39 ± 4 c	13.6 ± 1.2 c	0.147 ± 0.006	0.030 ± 0.002 b	0.016 ± 0.001 c	17.9 ± 1.5 b	4.0 ± 0.1 b
I×G	NS	NS	*	NS	NS	NS	*	*	NS

NS, *, **, and *** mean not significant and significant at 5%, 1%, and 0.1% by ANOVA. Different alphabet letters showed significance at 5% by the Tukey multiple comparison test. V% not detected. All parameters were $n = 5$, except for M% and A%, $n = 4$ and P% and P uptake, $n = 3$.

Table 2. Inoculation by genotype interaction (I×G) effect on root length density (RLD, cm/cm³) and shoot dry weight (SDW, g/plant) with control (C) and inoculants Dr. Kinkon (I₁) and Rootella P (I₂) and F (I₃) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 1).

Parameters	RLD				SDW			
	C	I ₁	I ₂	I ₃	C	I ₁	I ₂	I ₃
millet ₈₉₁	23 ± 4 abc	21 ± 3 abc	13 ± 3 ab	10 ± 2 a	0.007 ± 0.001 a	0.007 ± 0.001 a	0.007 ± 0.000 a	0.007 ± 0.001 a
millet ₉₅₄	22 ± 11 abc	10 ± 6 a	10 ± 4 a	14 ± 2 ab	0.004 ± 0.001 a	0.003 ± 0.001 a	0.004 ± 0.001 a	0.006 ± 0.00 a
rice _k	33 ± 5 c	33 ± 5 c	22 ± 6 bc	25 ± 3 bc	0.016 ± 0.003 bcd	0.016 ± 0.001 bcd	0.007 ± 0.002 a	0.014 ± 0.001 b
rice _t	33 ± 5 c	59 ± 4 d	33 ± 7 c	33 ± 4 c	0.018 ± 0.003 bcd	0.024 ± 0.002 e	0.015 ± 0.005 bc	0.020 ± 0.001 cde

Different alphabet letters show significance at 5% by least significant difference.

3.2. Exp. 2 Soil Types

UP exhibited a higher M% (32.9 vs. 10.2%) and P% (0.417 vs. 0.199%) than PD, while SDW and PH showed higher values in PD with a tendency for greater root growth parameters (RLD, RTA, and not significant) (Table 3). Millet₉₅₄, rice_k, and rice_t had a higher M% than millet₈₉₁, whereas shoot growth parameters and RLD were higher in rice than in

pearl millet. Rice (rice_t and rice_k) exhibited a higher SDW in PD, indicating a significant soil type by genotype interaction (SxG). An increased M% by inoculation led to a higher P uptake for millet₉₅₄ (Figure 1a) but not for rice (Figure 1b). Rice showed higher RLD (Figure 1d) than pearl millet (Figure 1c), and its higher RLD led to higher P uptake. P uptake in pearl millet increased by inoculation to a greater extent in UP and PD than in the other three inoculations and control (Figure 1a).

Table 3. Effects of soil type and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rate (%), root length density (RLD, cm/cm³), root transversal area (RTA, ×10⁴ μm²), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), plant height (PH, cm), and leaf age (LA) with indigenous AMF from Andosol upland soil (UP) and paddy soil (PD) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 2).

Parameters	M%	A%	RLD	RTA	P%	P Uptake	SDW	PH	LA
Soil (S)	**	NS	NS	NS	**	NS	***	***	***
UP	32.9 ± 4 b	2.5 ± 0.4	13 ± 4	6.9 ± 0.7	0.417 ± 0.020 b	0.015 ± 0.004	0.004 ± 0.002 a	7.4 ± 1.5 a	2.9 ± 0.1 a
PD	10.2 ± 6 a	0.4 ± 1.1	24 ± 2	8.4 ± 0.4	0.199 ± 0.085 a	0.021 ± 0.003	0.010 ± 0.000 b	11.5 ± 1.2 b	3.7 ± 0.1 b
Genotype (G)	***	NS	NS	NS	NS	*	***	***	***
millet ₈₉₁	4.2 ± 1.8 a	0.2 ± 0.2	12 ± 3	7.3 ± 0.9	0.249 ± 0.062	0.010 ± 0.002 a	0.005 ± 0.001 a	4.9 ± 0.8 a	3.0 ± 0.2 a
millet ₉₅₄	28.9 ± 8.1 b	2.6 ± 1.5	8 ± 2	8.2 ± 0.6	0.454 ± 0.136	0.012 ± 0.003 a	0.003 ± 0.000 a	3.5 ± 0.3 a	3.0 ± 0.1 a
rice _k	28.4 ± 10.9 b	2.8 ± 1.9	27 ± 5	5.6 ± 0.6	0.22 ± 0.020	0.022 ± 0.006 ab	0.009 ± 0.002 b	15.1 ± 1.2 b	3.5 ± 0.2 b
rice _t	24.7 ± 6.6 b	0.2 ± 0.2	27 ± 5	9.4 ± 0.9	0.31 ± 0.117	0.028 ± 0.005 b	0.011 ± 0.003 b	14.4 ± 1.6 b	3.7 ± 0.2 b
SxG	NS	NS	NS	NS	NS	NS	*	NS	NS

NS, *, **, and *** mean not significant and significant at 5%, 1%, and 0.1% by ANOVA. Different alphabet letters show significance at 5% by the Tukey multiple comparison test. V% not detected. All parameters were *n* = 4–5, except for P% and P uptake, *n* = 3.

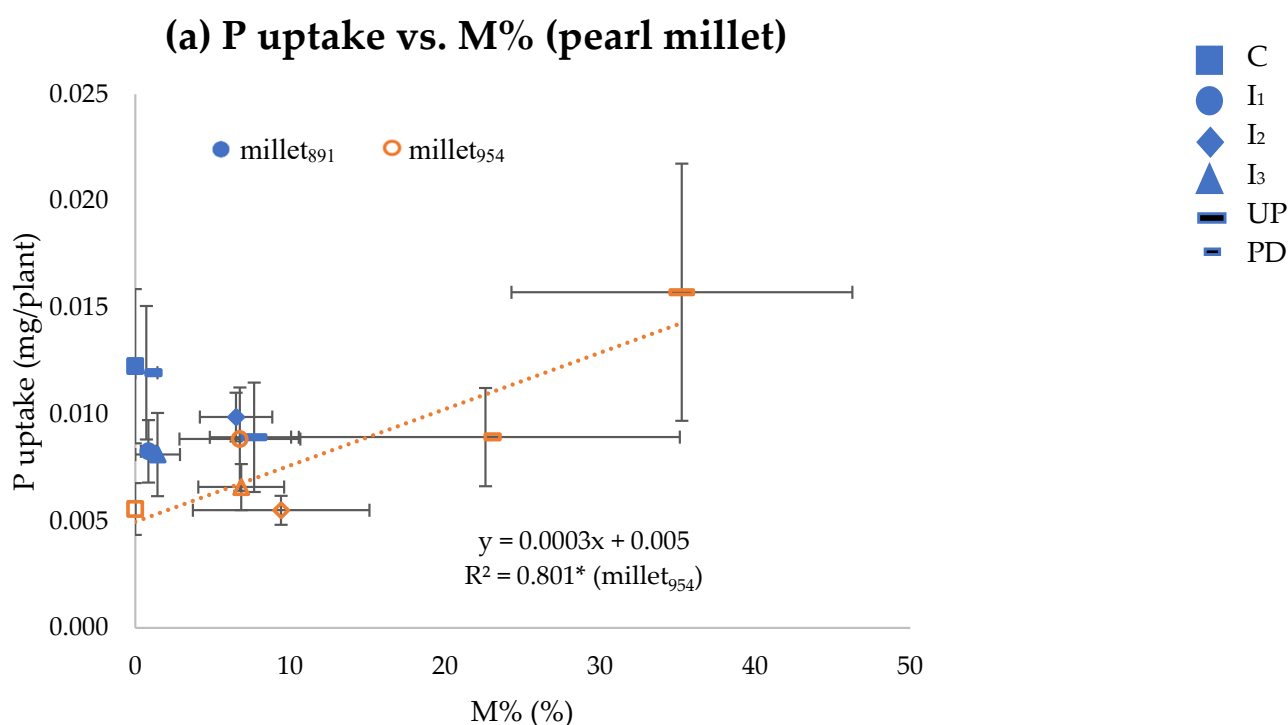


Figure 1. Cont.

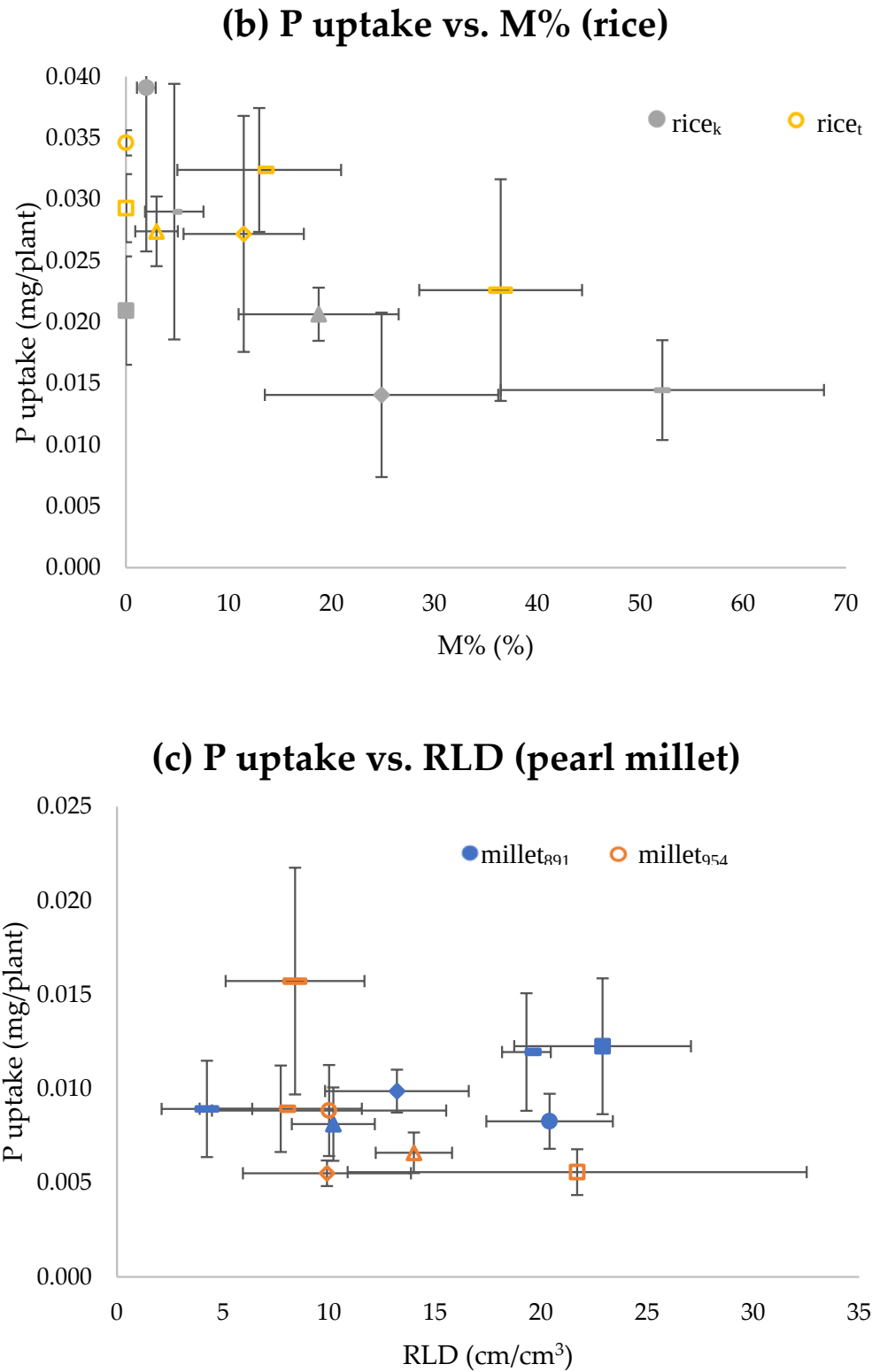


Figure 1. Cont.

(d) P uptake vs. RLD (rice)

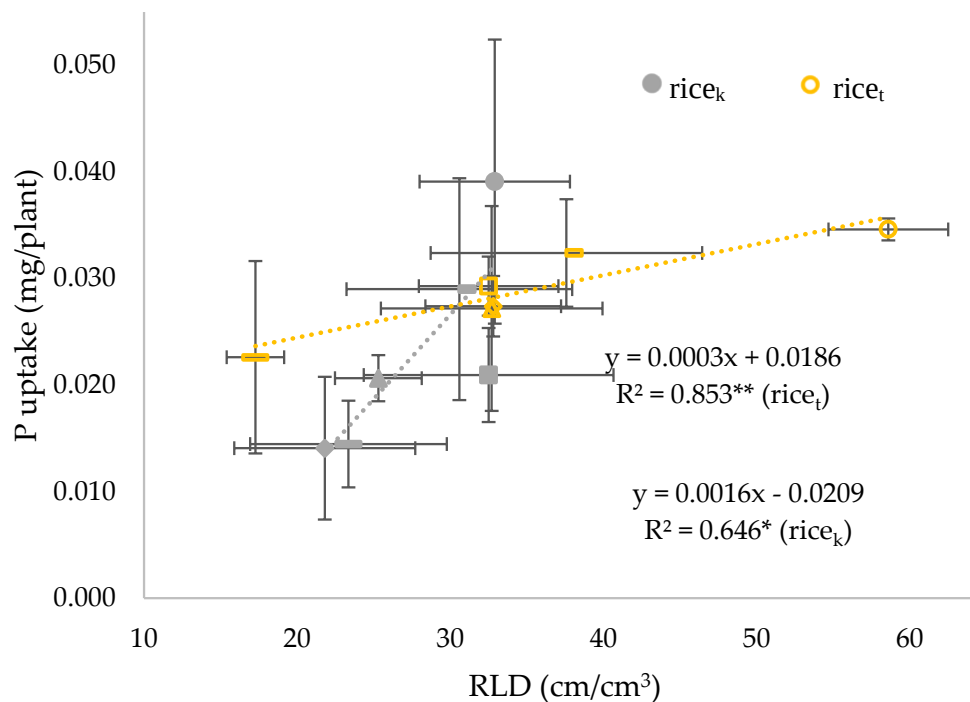


Figure 1. Relationships between mycorrhizal infection rate (M%) and P uptake, and root length density (RLD) and P uptake in Exp. 1 and 2 with control (C) and inoculants Dr. Kinkon (I₁) and Rootella P (I₂) and F (I₃), with indigenous AMF from Andosol upland soil (UP) and paddy soil (PD) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) (a,c) and rice (Koshihikari, rice_k; Togo4, rice_t) (b,d). Error bars indicate standard errors. Linear regressions are drawn when significant at $p < 0.05$ (*) and < 0.01 (**).

3.3. Exp. 3 and Exp. 4 Water Regimes

The four water regimes did not affect the infection rates (M% and A%) in Exp. 3, while SDW and PH decreased from FL to W100, W50, and W25; RLD and P uptake declined only under W25 (Table 4). P% was about 0.16% with no effects from the water regimes. Rice had a higher RLD and SDW than pearl millet. The five water regimes also did not affect the infection rates (M%, A%, and V%) in Exp. 4. However, shoot growth parameters (SDW, TN, and PH), P uptake, and RTA were ranked in the order of SS, FL, W100, W50, and W25 from highest to lowest values, while RLD was higher in W50 than FL and W25 (Table 5). P% was higher in W25 and W100 (ca. 0.14%) than SS (0.12%) and W50 (0.11%). Inoculation increased the M%, A%, RLD, RTA, SDW, and TN in Exp. 4. P%, P uptake, and SDW increased in inoculated pearl millet but not in inoculated rice (Table 6). Increasing the M% through inoculation resulted in greater P uptake for pearl millet (Figure 2a) but not for rice (Figure 2b), which exhibited a higher RLD (Figure 2c,d). The M% from inoculation was highest in W50 for pearl millet and W25 for rice, while P uptake from inoculation was highest in SS, followed by W100 in pearl millet and FL and SS in rice (Figure 2a,b).

Table 4. Effects of water and genotype on the mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rate (%), root length density (RLD, cm/cm³), root transversal area (RTA, $\times 10^4 \mu\text{m}^2$), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), and plant height (PH, cm) with inoculant Dr. Kinkon (I₁) in four water availabilities (flooded, FL; well irrigated, W100; 50% well irrigated, W50; 25% well irrigated, W25) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 3).

Parameters	M%	A%	RLD	RTA	P%	P Uptake	SDW	PH
Water (W)	NS	NS	***	***	NS	**	***	***
FL	1.6 ± 0.9	0.0 ± 0.0	27 ± 3 b	12.5 ± 1.2 b	0.153 ± 0.008	0.023 ± 0.004 b	0.015 ± 0.002 c	13.9 ± 1.8 b
W100	2.4 ± 1.1	0.0 ± 0.0	30 ± 5 b	12.5 ± 1.4 b	0.173 ± 0.027	0.023 ± 0.005 b	0.013 ± 0.002 bc	12.9 ± 1.7 ab
W50	0.5 ± 1.0	0.0 ± 0.6	30 ± 2 b	10.5 ± 0.6 b	0.166 ± 0.023	0.022 ± 0.003 b	0.011 ± 0.001 ab	12.6 ± 1.2 ab
W25	2.5 ± 0.4	0.5 ± 0.0	19 ± 3 a	6.2 ± 1.0 a	0.166 ± 0.011	0.011 ± 0.004 a	0.006 ± 0.001 a	10.7 ± 1.5 a
Genotype (G)	NS	NS	***	NS	NS	***	***	***
millet ₈₉₁	0.8 ± 0.5	0.0 ± 0.0	15 ± 2 a	10.1 ± 1.5	0.128 ± 0.016	0.008 ± 0.001 a	0.006 ± 0.001 a	6.4 ± 0.2 a
millet ₉₅₄	1.7 ± 1.2	0.0 ± 0.0	17 ± 2 a	9.1 ± 1.2	0.168 ± 0.015	0.009 ± 0.001 a	0.005 ± 0.000 a	5.7 ± 0.3 a
rice _k	1.9 ± 0.8	0.5 ± 0.6	35 ± 2 b	10.3 ± 1.0	0.184 ± 0.025	0.031 ± 0.004 b	0.016 ± 0.001 b	18.3 ± 0.7 b
rice _t	2.6 ± 1.0	0.0 ± 0.0	40 ± 3 c	12.2 ± 1.2	0.180 ± 0.013	0.032 ± 0.002 b	0.018 ± 0.001 b	19.7 ± 0.6 c
WxG	*	NS	***	NS	NS	NS	***	***

NS, *, **, and *** mean not significant and significant at 5%, 1%, and 0.1% by ANOVA. Different alphabet letters show significance at 5% by the Tukey multiple comparison test. V% was not detected. All parameters were $n = 5$, except for M% and A%, $n = 4$ and P% and P uptake, $n = 3$.

Table 5. Effects of water, inoculation, and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rate (%), root length density (RLD, cm/cm³), root transversal area (RTA, $\times 10^4 \mu\text{m}^2$), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), plant height (PH, cm), and tiller number (TN) of control (C) and I₁ (I) in 5 water regimes (flooded, FL; soil saturated, SS; well irrigated, W100; 50% well irrigated, W50; 25% well irrigated, W25) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2021 (Exp. 4).

Parameters	M%	A%	V%	RLD	RTA	P%	P Uptake	SDW	PH	TN
Water (W)	NS	NS	NS	***	***	**	***	***	***	***
FL	0.6 ± 0.2	0.0 ± 0.0	0.0 ± 0.0	9 ± 1 a	87.4 ± 6.4 d	0.127 ± 0.007 ab	0.354 ± 0.062 c	0.252 ± 0.037 d	30.5 ± 2.8 d	2.3 ± 0.2 ab
SS	1.3 ± 0.6	0.1 ± 0.1	0.1 ± 0.1	10 ± 1 ab	101.8 ± 3.7 e	0.118 ± 0.008 ab	0.439 ± 0.072 d	0.336 ± 0.043 e	30.8 ± 2.7 d	2.5 ± 0.3 b
W100	1.3 ± 0.4	0.1 ± 0.1	0.0 ± 0.0	10 ± 1 ab	77.0 ± 3.3 c	0.136 ± 0.01 ab	0.257 ± 0.028 b	0.206 ± 0.021 c	27.5 ± 1.9 c	2.0 ± 0.2 ab
W50	2.0 ± 0.4	0.1 ± 0.0	0.1 ± 0.1	11 ± 1 b	55.8 ± 2.7 b	0.112 ± 0.01 a	0.194 ± 0.022 ab	0.163 ± 0.010 b	24.8 ± 1.4 b	1.7 ± 0.1 ab
W25	1.6 ± 0.6	0.0 ± 0.1	0.1 ± 0.1	9 ± 1 a	40.5 ± 3.1 a	0.142 ± 0.005 b	0.163 ± 0.019 a	0.108 ± 0.012 a	19.1 ± 1.4 a	1.4 ± 0.1 a
Inoculation (I)	***	*	NS	***	***	NS	NS	***	NS	***
C	0.0 ± 0.0 a	0.0 ± 0.0 a	0.0 ± 0.0	9 ± 1 a	67.4 ± 3.8 a	0.124 ± 0.005	0.274 ± 0.035	0.198 ± 0.020 a	26.5 ± 1.4	1.9 ± 0.1 a
I	2.7 ± 0.4 b	0.1 ± 0.0 b	0.1 ± 0.0	10 ± 1 b	77.6 ± 3.6 b	0.130 ± 0.005	0.289 ± 0.028	0.228 ± 0.019 b	26.6 ± 1.4	2.0 ± 0.1 b
Genotype (G)	NS	NS	NS	***	***	NS	***	***	***	***
millet ₈₉₁	1.1 ± 0.4	0.0 ± 0.0	0.0 ± 0.0	6 ± 0 a	59.9 ± 3.9 a	0.122 ± 0.009	0.119 ± 0.013 b	0.096 ± 0.009 a	15.4 ± 0.5 a	2.0 ± 0.0 a
millet ₉₅₄	1.4 ± 0.4	0.1 ± 0.1	0.0 ± 0.0	6 ± 0 a	76.5 ± 5.4 b	0.120 ± 0.007	0.101 ± 0.013 a	0.079 ± 0.008 a	15.6 ± 0.5 a	2.0 ± 0.0 a
rice _k	1.6 ± 0.5	0.1 ± 0.0	0.1 ± 0.0	14 ± 0 b	62.5 ± 3.3 a	0.133 ± 0.009	0.440 ± 0.044 c	0.339 ± 0.025 b	38.1 ± 1.2 b	3.0 ± 0.1 b
rice _t	1.3 ± 0.4	0.0 ± 0.0	0.1 ± 0.0	14 ± 0 b	91.2 ± 6.5 c	0.134 ± 0.005	0.465 ± 0.043 d	0.339 ± 0.026 b	37.2 ± 1.2 b	3.0 ± 0.1 b
WxI	NS	NS	NS	NS	NS	***	NS	NS	***	NS
WxG	NS	NS	NS	NS	NS	**	***	***	***	***
IxG	NS	NS	NS	NS	NS	***	***	*	NS	*
WxIxG	NS	NS	NS	NS	NS	NS	*	NS	***	NS

NS, *, **, and *** mean not significant and significant at 5%, 1%, and 0.1% by ANOVA. Different alphabet letters show significance at 5% by the Tukey multiple comparison test. All parameters were $n = 4$, except for P% and P uptake, $n = 3$.

Table 6. Effects of IxG on shoot P concentration (P%, %), P uptake (mg/plant), and shoot dry weight (SDW, g/plant) with control (C) and I₁ (I) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2021 (Exp. 4).

Parameters	P%		P uptake		SDW	
	C	I	C	I	C	I
millet ₈₉₁	0.098 ± 0.008 a	0.145 ± 0.014 c	0.074 ± 0.014 a	0.163 ± 0.016 b	0.071 ± 0.012 a	0.120 ± 0.011 b
millet ₉₅₄	0.100 ± 0.006 a	0.140 ± 0.010 b	0.053 ± 0.009 a	0.150 ± 0.018 b	0.047 ± 0.007 a	0.111 ± 0.009 b
rice _k	0.150 ± 0.013 c	0.117 ± 0.009 ab	0.488 ± 0.063 d	0.392 ± 0.061 c	0.339 ± 0.034 c	0.339 ± 0.038 c
rice _t	0.145 ± 0.006 c	0.120 ± 0.005 ab	0.480 ± 0.062 d	0.451 ± 0.061 cd	0.334 ± 0.036 c	0.343 ± 0.037 c

Different alphabet letters show significance at 5% by the least square significance test.

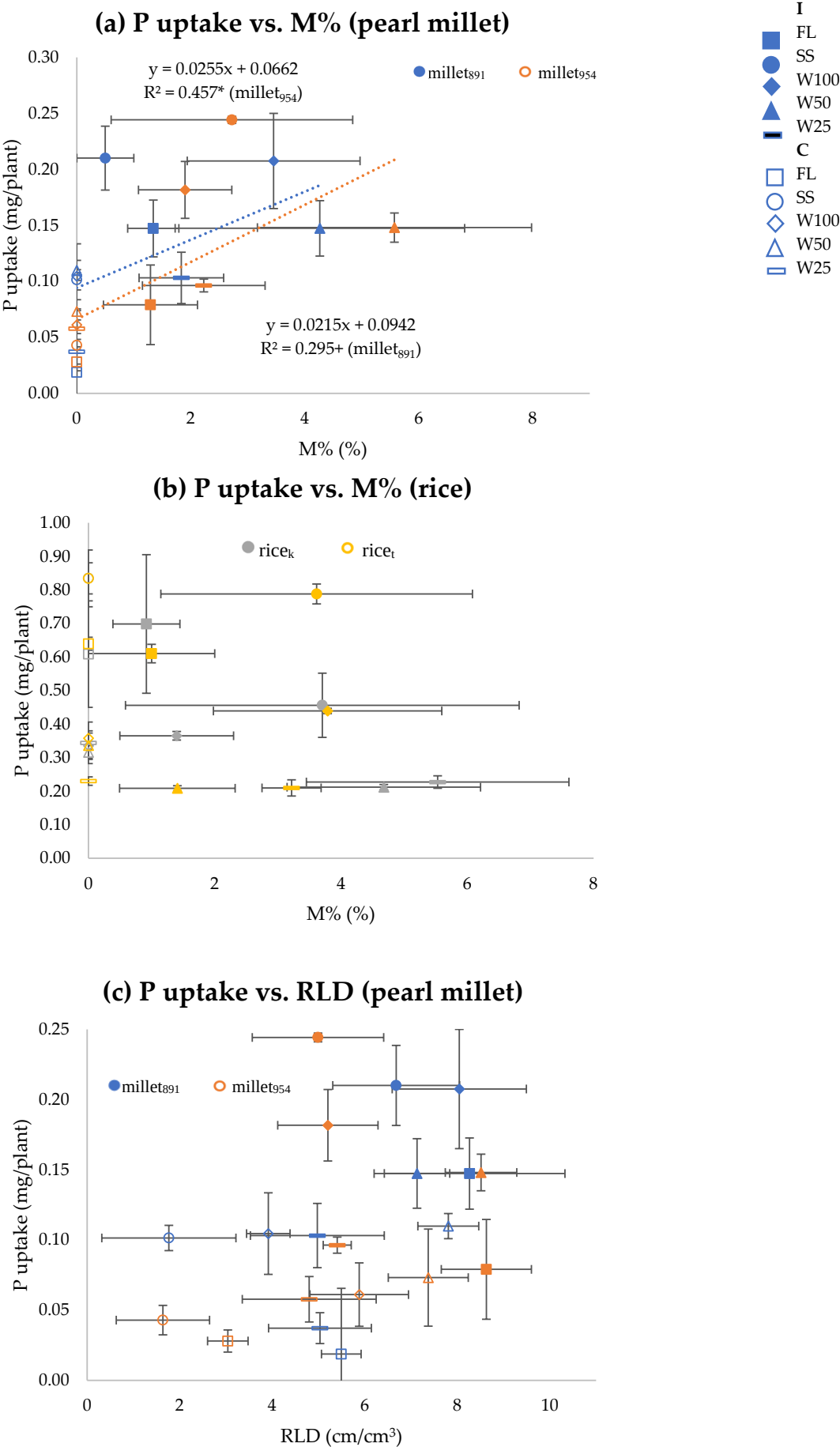


Figure 2. Cont.

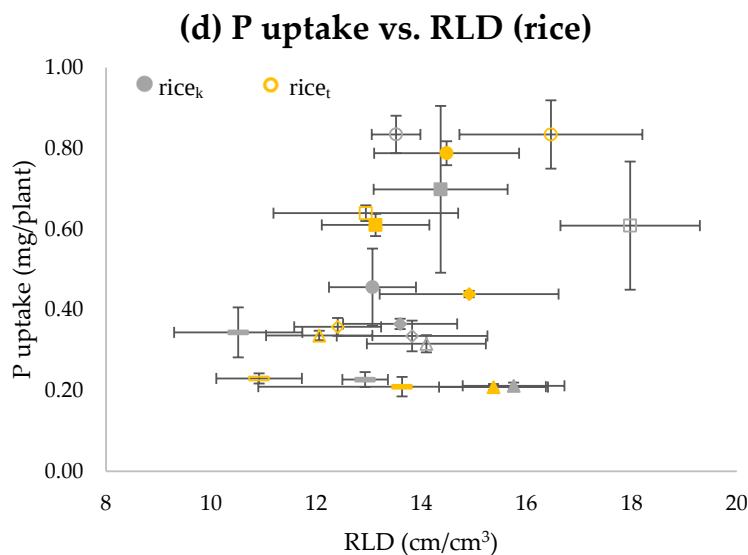


Figure 2. Relationships between mycorrhizal infection rate (M%) and P uptake, and root length density (RLD) and P uptake in Exp. 4 for pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) (a,c) and rice (Koshihikari, rice_k; Togo4, rice_t) (b,d) with/without inoculation (I, C) across five water regimes (flooded, FL; saturated soil, SS; well irrigated, W100; 50% well irrigated, W50; 25% well irrigated, W25). Error bars indicated standard errors. Linear regressions are drawn when significant at $p < 0.05$ (*).

4. Discussion

4.1. Inoculation Types and Indigenous AMF

This study showed small but positive effects of all three inoculants on either infection rates or plant growth parameters with some unique differences between the new inoculants, Rootella (I₂ and I₃), and the popular Japanese inoculant Dr Kinkon (I₁). The common characteristics of all three inoculants included an enhancement in the infection rate of 3-week-old rice and pearl millet seedlings, as well as a slower development (i.e., smaller LA) compared to the control. The new inoculant, Rootella, exhibited a higher M% and smaller LA (Table 1). As part of the assimilates need to be competitively provided to AMF colonization, shoot development might have been delayed in well-infected seedlings [20], as seen from the delayed leaf development. Higher infection rates (i.e., M%) of Rootella would be due to their higher propagule numbers, as indicated by Karasawa et al. [21], but this infection rate was not directly translated into plant growth in the new inoculants. Only the inoculant Dr. Kinkon I₁ increased the RTA, SDW, and PH more than the control, whereas Rootella did not show such effects, but I₂ showed the suppressed RLD and SDW. The I₁ is still uncertain as the carriers attached to I₁, like crystalline silica, may or may not have enhanced biomass production, and silica nanoparticles are reported to promote plant growth and improve plant resistance against biotic and abiotic stresses [22]. However, Ortas and Akpınar [23] showed an enhancement of spore production (695 vs. 0 per 100 g soil) and M% (84 vs. 30%) by I₁ inoculation over control on six maize genotypes in the pots at the seeding stage at eight weeks, which was comparable or higher than other mycorrhizal inoculations (seven glomus species, one indigenous mycorrhizae, and their cocktail). I₁ did not change the shoot dry weight but enhanced the root dry weight and plant concentrations of phosphorus and zinc. Nevertheless, Niwa et al. [15] argued that Dr. Kinkon I₁ had typical ruderal traits, which could lead to early infection with mass spore production and good adaptation to agricultural land. Thus, further studies are needed under field conditions to assess the full potential of the new inoculants Rootella with higher propagule density under field conditions since the low infection rates of the current study may have been

related to the relatively small volume of the soils in the cell tray. Also, caution is needed when reporting the effects of the commercial mycorrhizal inoculants. From the current inoculation treatment (14 g per tray), the estimated number of propagules could range from 196 (I_1) to 350,000 (I_3) propagules per tray, according to the product's explanation; however, the actual responding numbers from the internal observation were significantly lower. Salmon et al. [14] also warned that the number of commercial mycorrhizal inoculants did not positively impact root colonization or plant growth.

This study also confirmed that natural soils can become a substantial source of mycorrhizal infection, as was shown by a much higher infection rate from the indigenous AMF collected from the topsoil of Andosol than the autoclaved soils added with the inoculants (Tables 1 and 3; Figure 1). The higher infection rate led to higher P uptake found in millet⁹⁵⁴. It is well known that lowland rice can be infected with AMF but with much lower infection rates than upland rice [24,25], but it was surprising that the natural soils collected from lowland fields also showed substantial infection rates in this experiment. This may be because the soils in the lowland field in this experiment had been fallowed under aerobic conditions from autumn to spring of the previous year when substantial amounts of weeds were infested. However, the soils from the upland field exhibited significantly higher spore and propagule counts of indigenous AMF, as AMF infection measured by qPCR in the upland Andosol was 50% greater than in the lowland Andosol at the same experimental site (K Ejiri, personal communication). From this observation, using indigenous AMF may be one good inoculation source, in addition to using new commercial inoculants.

4.2. Inoculation and Water Regime Effects

This study showed that the P uptake of pearl millet seedlings was more responsive to I_1 inoculant than rice seedlings (Figure 2a,b; Tables 5 and 6). The superior response of pearl millet to rice was also shown in the Andosol field experiment [26]. Inoculation improved shoot biomass from higher P uptake in maize at 10 weeks after sowing with 0, 10, and 50 g of I_1 in 2 kg of dry Andosol, and the relative effect was more noticeable under a low water regime due to the effect of P uptake on growth [21]. Under flooded (FL) conditions, the water regime may limit the infection rates by directly influencing aerobic AMF or increasing root aerenchyma [27]. This study found that rice did not increase phosphorus (P) uptake or shoot dry weight (SDW) through inoculation. It is possible that rice could absorb P directly due to a better-developed root system from the nursery soil.

The infection rates, measured as M%, A%, and V% through microscopic observation, were generally unaffected by the water regimes in our cell tray study. This finding was contrary to our expectation based on previous studies that indicated higher infection rates under water-limiting conditions than those with ample water supply (e.g., [5]). A similar negligible response in infection rates to water regimes was obtained in the same set of pearl millet and rice genotypes under Andosols in the field experiments [24,26]. The discrepancy between this result and the expectation might be related to the magnitude of changes in rhizosphere P availability when water availability changes, which needs further experimental proof. The small response in our cell tray experiments might be partially due to the low infection rates with the commercial soil media [28] and the small volume of soil in the cell tray. Water regimes did change the composition of AMF species in rice and pearl millet [24,25], but the overall infection rates did not dramatically change. Although this experiment cannot strongly prove it, water limitation (e.g., W25) may reduce carbon assimilation, make roots shorter and thinner, and induce complicated changes that influence mycorrhizal infection.

5. Conclusions

The new inoculants, Rootella (I₂ and I₃), with higher propagule numbers, showed a higher infection rate than the control seedlings in both rice and pearl millet, with the tendency for slower leaf development and no seedling growth enhancement. The popular Japanese inoculant Dr. Kinkon (I₁) had more positive effects on the root transversal area and shoot growth parameters than the control. The infection rates of all three inoculants were lower than the indigenous AMF from upland Andosol in rice and pearl millet, in which a higher infection rate led to a higher P uptake found in millet₉₅₄. The inoculant Dr. Kinkon increased the infection rate in pearl millet and rice but did not indicate interaction with water regimes. A higher infection rate led to a higher P uptake and shoot dry weight in pearl millet but not in rice with higher root length density. This study provided the significance of inoculants for seedling establishment and highlighted more mycorrhizal-mediated P uptake in pearl millet than in rice.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/agriculture15070753/s1>, Figure S1: Mycorrhizal structures in infected roots dyed with trypan blue. Bar = 50 μ m. Table S1. Effects of inoculation and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rates in (%), root length density (RLD, cm/cm³), root transversal area (RTA, $\times 10^4 \mu$ m²), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), plant height (PH, cm), and leaf age (LA) with control (C) and inoculants Dr. Kinkon (I₁) and Rootella P (I₂) and F (I₃) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 1). Table S2. Effects of soil type and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rate (%), root length density (RLD, cm/cm³), root transversal area (RTA, $\times 10^4 \mu$ m²), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), plant height (PH, cm), and leaf age (LA) with indigenous AMF from Andosol upland soil (UP) and paddy soil (PD) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 2). Table S3. Effects of water and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rate (%), root length density (RLD, cm/cm³), root transversal area (RTA, $\times 10^4 \mu$ m²), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), and plant height (PH, cm) with inoculant Dr. Kinkon (I₁) in four water availabilities (flooded, FL; well irrigated, W100; 50% well irrigated, W50; 25% well irrigated, W25) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2020 (Exp. 3). Table S4. Effects of water, inoculation, and genotype on mycorrhizal (M%), arbuscular (A%), and vesicular (V%) infection rate (%), root length density (RLD, cm/cm³), root transversal area (RTA, $\times 10^4 \mu$ m²), shoot P concentration (P%, %) and P uptake (mg/plant), shoot dry weight (SDW, g/plant), plant height (PH, cm), and tiller number (TN) of control (C) and I₁ (I) in 5 water regimes (flooded, FL; soil saturated, SS; well irrigated, W100; 50% well irrigated, W50; 25% well irrigated, W25) in pearl millet (ICMB89111, millet₈₉₁; ICMB95444, millet₉₅₄) and rice (Koshihikari, rice_k; Togo4, rice_t) in 2021 (Exp. 4).

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Data Availability Statement: Publicly available datasets were analyzed in this study. These data can be found here: [<https://1drv.ms/f/s!Ah0xdLQv16t7hoN17rcxYG3qN7LnmQ?e=Edj45F>] (accessed on 19 March 2025).

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